

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and data scientist with experience developing and validating statistical and machine-learning solutions for financial risk, insurance pricing, and large, noisy datasets. Builds reproducible Python/PyTorch workflows spanning data preparation, feature processing, model training, validation, simulation, and monitoring diagnostics, and translates technical results into clear decisions for business stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Feb 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analytical pipelines

Data Science and Model Lifecycle: Data preparation, feature processing, supervised learning, GLM/GBM, regression, model selection and validation, performance evaluation, feature attribution, hypothesis testing, Monte Carlo simulation, numerical optimization, model diagnostics, documentation, and stakeholder communication

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring requirement into explicit hypotheses, likelihood-ratio boundaries, and controlled stopping decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, Type I/II error, stopping behavior, uncertainty, and model limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

RESEARCH AND ENGINEERING

Ph.D. Dissertation Research - Scalable Model Development and Statistical Diagnostics

Sep 2024 - Present

- Scalable Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- Data Quality and Root-Cause Analysis:** Built workflows for quality filtering, missingness tracking, time-specific coordinate offsets, nearest-neighbor center matching, and irregular-to-regular grid mapping to separate systematic acquisition artifacts from random noise and model misspecification.
- Model Evaluation and Monitoring:** Designed scale- and frequency-resolved diagnostics to localize whether model discrepancies arise from low-frequency structure, high-frequency behavior, noise, missingness, or dependence.
- Research Engineering:** Implemented reusable Python/PyTorch pipelines with CPU/GPU execution, cached artifacts, and restartable HPC jobs; integrated compiled C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Linear Algebra, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)